สูตรสำหรับสอบปลายภาค วิชาสนามแม่เหล็กไฟฟ้า 1
ปีการศึกษา 2569

ขนาดของเวกเตอร์

$$\mathbf{B} = B_x\hat{\mathbf{a}}_x + B_y\hat{\mathbf{a}}_y + B_z\hat{\mathbf{a}}_z,\ |\mathbf{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2},\ \hat{\mathbf{a}}_B = \frac{\mathbf{B}}{|\mathbf{B}|}$$

ผลคูณจุด (dot product)

$$\mathbf{A}\cdot\mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta_{AB}$$

$$\mathbf{A} = A_x\hat{\mathbf{a}}_x + A_y\hat{\mathbf{a}}_y + A_z\hat{\mathbf{a}}_z,\ \mathbf{A}\cdot\mathbf{B} = A_xB_x + A_yB_y + A_zB_z,\ \mathbf{A}\cdot\mathbf{A} = |\mathbf{A}|^2$$

ผลคูณไขว้ (cross product)

$$\mathbf{A}\times\mathbf{B} = \hat{\mathbf{a}}_N|\mathbf{A}||\mathbf{B}|\sin\theta_{AB}$$

$$\mathbf{A}\times\mathbf{B} = (A_yB_z - A_zB_y)\hat{\mathbf{a}}_x + (A_zB_x - A_xB_z)\hat{\mathbf{a}}_y + (A_xB_y - A_yB_x)\hat{\mathbf{a}}_z$$

$$\mathbf{A}\times\mathbf{B} = \begin{vmatrix} \hat{\mathbf{a}}_x & \hat{\mathbf{a}}_y & \hat{\mathbf{a}}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

ระบบพิกัดทรงกระบอก

$$x = \rho\cos\phi,\ y = \rho\sin\phi,\ z = z,\ \rho = \sqrt{x^2 + y^2},\ \phi = \tan^{-1}\frac{y}{x}$$

$$\begin{bmatrix} \hat{\mathbf{a}}_\rho \\ \hat{\mathbf{a}}_\phi \\ \hat{\mathbf{a}}_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{a}}_x \\ \hat{\mathbf{a}}_y \\ \hat{\mathbf{a}}_z \end{bmatrix},\ \begin{bmatrix} \hat{\mathbf{a}}_\rho \\ \hat{\mathbf{a}}_\phi \\ \hat{\mathbf{a}}_z \end{bmatrix} \equiv [A]_{rc} \begin{bmatrix} \hat{\mathbf{a}}_x \\ \hat{\mathbf{a}}_y \\ \hat{\mathbf{a}}_z \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{a}}_x \\ \hat{\mathbf{a}}_y \\ \hat{\mathbf{a}}_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{a}}_\rho \\ \hat{\mathbf{a}}_\phi \\ \hat{\mathbf{a}}_z \end{bmatrix},\ \begin{bmatrix} \hat{\mathbf{a}}_x \\ \hat{\mathbf{a}}_y \\ \hat{\mathbf{a}}_z \end{bmatrix} \equiv [A]_{cr} \begin{bmatrix} \hat{\mathbf{a}}_\rho \\ \hat{\mathbf{a}}_\phi \\ \hat{\mathbf{a}}_z \end{bmatrix}$$

$$[A]_{rc}^{-1} = [A]_{rc}^T,\ [A]_{cr} = [A]_{rc}^T$$

$$\mathbf{A} = A_x\hat{\mathbf{a}}_x + A_y\hat{\mathbf{a}}_y + A_z\hat{\mathbf{a}}_z,\ \mathbf{A} = A_\rho\hat{\mathbf{a}}_\rho + A_\phi\hat{\mathbf{a}}_\phi + A_z\hat{\mathbf{a}}_z$$

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix},\ \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} \equiv [A]_{rc} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}, \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} \equiv [A]_{cr} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

ระบบพิกัดทรงกลม

$$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta$$

$$r = \sqrt{x^2 + y^2 + z^2},\ \theta = \cos^{-1}\frac{z}{\sqrt{x^2 + y^2 + z^2}},\ \phi = \tan^{-1}\frac{y}{x}$$

$$\begin{bmatrix} \hat{\mathbf{a}}_r \\ \hat{\mathbf{a}}_\theta \\ \hat{\mathbf{a}}_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{a}}_x \\ \hat{\mathbf{a}}_y \\ \hat{\mathbf{a}}_z \end{bmatrix}, \begin{bmatrix} \hat{\mathbf{a}}_r \\ \hat{\mathbf{a}}_\theta \\ \hat{\mathbf{a}}_\phi \end{bmatrix} \equiv [A]_{rs} \begin{bmatrix} \hat{\mathbf{a}}_x \\ \hat{\mathbf{a}}_y \\ \hat{\mathbf{a}}_z \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{a}}_x \\ \hat{\mathbf{a}}_y \\ \hat{\mathbf{a}}_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{a}}_r \\ \hat{\mathbf{a}}_\theta \\ \hat{\mathbf{a}}_\phi \end{bmatrix}, \begin{bmatrix} \hat{\mathbf{a}}_x \\ \hat{\mathbf{a}}_y \\ \hat{\mathbf{a}}_z \end{bmatrix} \equiv [A]_{sr} \begin{bmatrix} \hat{\mathbf{a}}_r \\ \hat{\mathbf{a}}_\theta \\ \hat{\mathbf{a}}_\phi \end{bmatrix}$$

$$[A]_{rs}^{-1} = [A]_{rs}^T,\ [A]_{sr} = [A]_{rs}^T$$

$$\mathbf{A} = A_x\hat{\mathbf{a}}_x + A_y\hat{\mathbf{a}}_y + A_z\hat{\mathbf{a}}_z,\ \mathbf{A} = A_r\hat{\mathbf{a}}_r + A_\theta\hat{\mathbf{a}}_\theta + A_\phi\hat{\mathbf{a}}_\phi$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}, \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} \equiv [A]_{rs} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}, \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} \equiv [A]_{sr} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

กฎของคูลอมบ์ (Coulomb's law)

$$\mathbf{F}_2 = k\frac{Q_1Q_2}{|\mathbf{R}_{12}|^2}\hat{\mathbf{a}}_{12},\ \mathbf{R}_{12} = -\mathbf{r}_1 + \mathbf{r}_2 = \mathbf{r}_2 - \mathbf{r}_1,\ \hat{\mathbf{a}}_{12} = \frac{\mathbf{R}_{12}}{|\mathbf{R}_{12}|}, \mathbf{F}_1 = k\frac{Q_2Q_1}{|\mathbf{R}_{21}|^2}\hat{\mathbf{a}}_{21}$$

$$k = \frac{1}{4\pi\varepsilon_0},\ \varepsilon_0 = 8.854\times10^{-12}\ \text{F/m}$$

สนามไฟฟ้าเนื่องจากประจุแบบจุด

$$\mathbf{E} = k\frac{Q}{|\mathbf{R}|^2}\hat{\mathbf{a}}_R,\ \mathbf{R} = \mathbf{r} - \mathbf{r}',\ \hat{\mathbf{a}}_R = \frac{\mathbf{R}}{|\mathbf{R}|}$$

สนามไฟฟ้าเนื่องจากประจุแบบจุดหลายตัว

$$\mathbf{E} = \sum_{i=1}^{N} k \frac{Q_i}{\left|\mathbf{R}_i\right|^2} \hat{\mathbf{a}}_{R,i}$$

การกระจายของประจุ และความหนาแน่นประจุ

$$\rho_l = \lim_{\Delta l \to 0} \frac{\Delta Q}{\Delta l},\ \rho_s = \lim_{\Delta s \to 0} \frac{\Delta Q}{\Delta s},\ \rho_v = \lim_{\Delta v \to 0} \frac{\Delta Q}{\Delta v}$$

สนามไฟฟ้าเนื่องจากการกระจายของประจุ

$$d\mathbf{E} = \hat{\mathbf{a}}_R k \frac{dQ}{\left|\mathbf{R}\right|^2},\ dQ = \rho_l dl',\ \mathbf{R} = -\mathbf{r}' + \mathbf{r} = \mathbf{r} - \mathbf{r}',\ \hat{\mathbf{a}}_R = \frac{\mathbf{R}}{\left|\mathbf{R}\right|}$$

$$\int \cos\psi d\psi = \sin\psi,\ \frac{d\tan\psi}{d\psi} = \sec^2\psi$$

$$\mathbf{E} = \int_S \hat{\mathbf{a}}_R k \frac{\rho_s}{\left|\mathbf{R}\right|^2} ds',\ \mathbf{E} = \int_V \hat{\mathbf{a}}_R k \frac{\rho_v}{\left|\mathbf{R}\right|^2} dv'$$

ส่วนย่อยปริมาณน้อย ๆ ของพื้นที่ และปริมาตร

$dv = dxdydz,\ ds = dxdy,\ ds = dydz,\ ds = dxdz$ ระบบพิกัดฉาก

$dv = \rho d\rho d\phi dz,\ ds = \rho d\rho d\phi,\ ds = d\rho dz,\ ds = \rho d\phi dz$ ระบบพิกัดทรงกระบอก

$dv = r^2 \sin\theta dr d\theta d\phi,\ ds = r\sin\theta dr d\phi,\ ds = r dr d\theta,\ ds = r^2 \sin\theta d\theta d\phi$ ระบบพิกัดทรงกลม

ฟลักซ์ไฟฟ้า และความหนาแน่นฟลักซ์

$$\Psi = Q,\ \mathbf{D} = \frac{\Psi}{4\pi r^2}\hat{\mathbf{a}}_r = \frac{Q}{4\pi r^2}\hat{\mathbf{a}}_r$$

กฎของเกาส์ (Gauss's law)

$$\Psi = Q,\ \Psi = \oint_S \mathbf{D}_s \cdot d\mathbf{s},\ \Psi = \oint_S \mathbf{D}_s \cdot d\mathbf{s} = Q$$

การประยุกต์กฎของเกาส์และไดเวอร์เจนซ์

$$\mathbf{D}_0 = D_{x0}\hat{\mathbf{a}}_x + D_{y0}\hat{\mathbf{a}}_y + D_{z0}\hat{\mathbf{a}}_z$$

$$\oint_S \mathbf{D}\cdot d\mathbf{s} = Q,\ \oint_S \mathbf{D}\cdot d\mathbf{s} = \int_{\text{front}} + \int_{\text{back}} + \int_{\text{left}} + \int_{\text{right}} + \int_{\text{top}} + \int_{\text{bottom}}$$

$$\int_{\text{front}} = \mathbf{D}_{\text{front}} \cdot \Delta\mathbf{s}_{\text{front}}$$

$$\mathbf{D}_{\text{front}} = D_{x,\text{front}}\hat{\mathbf{a}}_x + D_{y,\text{front}}\hat{\mathbf{a}}_y + D_{z,\text{front}}\hat{\mathbf{a}}_z,\ \Delta\mathbf{s}_{\text{front}} = \Delta y \Delta z \hat{\mathbf{a}}_x$$

$$\int_{\text{front}} + \int_{\text{back}} = \frac{\partial D_x}{\partial x}\Delta x \Delta y \Delta z$$

$$\nabla \cdot \mathbf{D} = \rho_v$$

ไดเวอร์เจนซ์ (divergence)

$$\nabla \cdot \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial \mathrm{z}}$$ ระบบพิกัดฉาก

$$\nabla \cdot \mathbf{D} = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho D_\rho) + \frac{1}{\rho}\frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial \mathrm{z}}$$ ระบบพิกัดทรงกระบอก

$$\nabla \cdot \mathbf{D} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 D_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(D_\theta \sin\theta) + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial \phi}$$

ระบบพิกัดทรงกลม

พลังงานที่ใช้ในการเคลื่อนประจุแบบจุดท่ามกลางสนามไฟฟ้า

$$\mathbf{F}_E = \mathbf{E}Q,\ \mathbf{F}_{\text{appl}} = -\mathbf{F}_E = -\mathbf{E}Q,\ dW = \mathbf{F}_{\text{appl}} \cdot d\mathbf{l} = -\mathbf{E}Q \cdot d\mathbf{l},\ W = -Q\int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{l}$$

$$\frac{W}{Q} = -\int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{l},\ \text{potential difference} = -\int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{l}$$

$$V_{AB} = -\int_{B}^{A} \mathbf{E} \cdot d\mathbf{l}$$

$$d\mathbf{l} = dx\hat{\mathbf{a}}_x + dy\hat{\mathbf{a}}_y + dz\hat{\mathbf{a}}_z$$ ระบบพิกัดฉาก

$$d\mathbf{l} = d\rho\hat{\mathbf{a}}_\rho + \rho d\phi\hat{\mathbf{a}}_\phi + dz\hat{\mathbf{a}}_z$$ ระบบพิกัดทรงกระบอก

$$d\mathbf{l} = dr\hat{\mathbf{a}}_r + rd\theta\hat{\mathbf{a}}_\theta + r\sin\theta d\phi\hat{\mathbf{a}}_\phi$$ ระบบพิกัดทรงกลม

ศักย์ไฟฟ้า (potential)

$$V_{AB} = V_A - V_B,\ V \equiv V_A = V_{AB}$$

การหาสนามไฟฟ้าในจากเกรเดียนต์ (gradient)

$$\mathbf{E} = -\nabla V$$

$$\nabla V = \frac{\partial V}{\partial x}\hat{\mathbf{a}}_x + \frac{\partial V}{\partial y}\hat{\mathbf{a}}_y + \frac{\partial V}{\partial \mathrm{z}}\hat{\mathbf{a}}_z$$ ระบบพิกัดฉาก

$$\nabla V = \frac{\partial V}{\partial \rho}\hat{\mathbf{a}}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\hat{\mathbf{a}}_\phi + \frac{\partial V}{\partial \mathrm{z}}\hat{\mathbf{a}}_z$$ ระบบพิกัดทรงกระบอก

$$\nabla V = \frac{\partial V}{\partial r}\hat{\mathbf{a}}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{\mathbf{a}}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\hat{\mathbf{a}}_\phi$$ ระบบพิกัดทรงกลม

กระแสไฟฟ้า และความหนาแน่นกระแสไฟฟ้า (electric current and electric current density)

$$I = \frac{dQ}{dt},\ \Delta I = \mathbf{J}\cdot\Delta\mathbf{s},\ I = \int_S \mathbf{J}\cdot d\mathbf{s}$$

ความหนาแน่นกระแสไฟฟ้าการพา (convection electric current density)

$$\Delta I = \frac{\Delta Q}{\Delta t},\ \Delta Q = \rho_v \Delta v = \rho_v(\Delta s \Delta x)$$

$$\Delta I = \frac{\Delta Q}{\Delta t} = \rho_v \Delta s \frac{\Delta x}{\Delta t},\ \Delta I = \rho_v \Delta s v_x,\ J_x = \frac{\Delta I}{\Delta s} = \frac{\rho_v \Delta s v_x}{\Delta s} = \rho_v v_x,\ \mathbf{J} = \rho_v \mathbf{v}$$

ความต่อเนื่องของกระแส (continuity of current)

$$I = -\frac{dQ_i}{dt},\ \oint_S \mathbf{J}\cdot d\mathbf{s} = -\frac{dQ_i}{dt},\ \oint_S \mathbf{J}\cdot d\mathbf{s} = -\int_{\text{vol}} \frac{\partial \rho_v}{\partial t} dv,\ \oint_S \mathbf{J}\cdot d\mathbf{s} = \int_{\text{vol}} (\nabla\cdot\mathbf{J}) dv$$

$$\int_{\text{vol}} (\nabla\cdot\mathbf{J}) dv = -\int_{\text{vol}} \frac{\partial \rho_v}{\partial t} dv,\ \nabla\cdot\mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

ไดเวอร์เจนซ์

$$\nabla\cdot\mathbf{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}$$ ระบบพิกัดฉาก

$$\nabla\cdot\mathbf{J} = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho J_\rho) + \frac{1}{\rho}\frac{\partial J_\phi}{\partial \phi} + \frac{\partial J_z}{\partial z}$$ ระบบพิกัดทรงกระบอก

$$\nabla\cdot\mathbf{J} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 J_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(J_\theta \sin\theta) + \frac{1}{r\sin\theta}\frac{\partial J_\phi}{\partial \phi}$$ ระบบพิกัดทรงกลม

ธรรมชาติของไดอิเล็กทริก (nature of dielectric)

$$\varepsilon = \varepsilon_r \varepsilon_0$$

เงื่อนไขขอบเขต (boundary condition)

$$\oint_C \mathbf{E}\cdot d\mathbf{l} = 0,\ \oint_C \mathbf{E}\cdot d\mathbf{l} = \left(\int_1 + \int_2 + \int_3 + \int_4\right)\mathbf{E}\cdot d\mathbf{l} = 0,\ E_{\tan 1} = E_{\tan 2}$$

$$\oint_S \mathbf{D}\cdot d\mathbf{s} = Q,\ \oint_S \mathbf{D}\cdot d\mathbf{s} = \left(\int_{\text{top}} + \int_{\text{bottom}} + \int_{\text{side}}\right)\mathbf{D}\cdot d\mathbf{s} = \Delta Q,\ D_{N1} - D_{N2} = \rho_s$$

ความจุไฟฟ้า (capacitance)

$$C = \frac{Q}{V_0}$$

สมการของปัวซงส์ (Poisson's equation) และสมการของลาปลาซ (Laplace's equation)

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon},\ \nabla^2 V = 0$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

ระบบพิกัดฉาก

$$\nabla^2 V = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\left(\frac{\partial^2 V}{\partial \phi^2}\right) + \frac{\partial^2 V}{\partial z^2}$$

ระบบพิกัดทรงกระบอก

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 V}{\partial \phi^2}$$

ระบบพิกัดทรงกลม

สนามไฟฟ้าในรูปเกรเดียนต์ (gradient)

$$\mathbf{E} = -\nabla V$$

$$\nabla V = \frac{\partial V}{\partial x}\hat{\mathbf{a}}_x + \frac{\partial V}{\partial y}\hat{\mathbf{a}}_y + \frac{\partial V}{\partial z}\hat{\mathbf{a}}_z$$ ระบบพิกัดฉาก

$$\nabla V = \frac{\partial V}{\partial \rho}\hat{\mathbf{a}}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\hat{\mathbf{a}}_\phi + \frac{\partial V}{\partial z}\hat{\mathbf{a}}_z$$ ระบบพิกัดทรงกระบอก

$$\nabla V = \frac{\partial V}{\partial r}\hat{\mathbf{a}}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{\mathbf{a}}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\hat{\mathbf{a}}_\phi$$ ระบบพิกัดทรงกลม

กฎบิโอต์-ซาวารต์ (Biot-Savart law)

$$d\mathbf{H} = \frac{Id\mathbf{l}' \times \hat{\mathbf{a}}_R}{4\pi|\mathbf{R}|^2},\ \mathbf{R} = \mathbf{r} - \mathbf{r}',\ \mathbf{H} = \int_S \frac{\mathbf{K} \times \hat{\mathbf{a}}_R ds'}{4\pi|\mathbf{R}|^2},\ \mathbf{H} = \int_{\text{vol}} \frac{\mathbf{J} \times \hat{\mathbf{a}}_R dv'}{4\pi|\mathbf{R}|^2}$$

กฎวงจรของแอมแปร์ (Ampere's circuital law)

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I$$

เคิร์ล (curl)

$$\mathbf{H}_0 = H_{x0}\hat{\mathbf{a}}_x + H_{y0}\hat{\mathbf{a}}_y + H_{z0}\hat{\mathbf{a}}_z,\ \oint_C \mathbf{H} \cdot d\mathbf{l} = I,\ \oint_C \mathbf{H} \cdot d\mathbf{l} = \int_{1-2} + \int_{2-3} + \int_{3-4} + \int_{4-1}$$

$$\int_{1-2} = \mathbf{H}_{1-2} \cdot \Delta\mathbf{l}_{1-2},\ \mathbf{H}_{1-2} = H_{x,1-2}\hat{\mathbf{a}}_x + H_{y,1-2}\hat{\mathbf{a}}_y + H_{z,1-2}\hat{\mathbf{a}}_z,\ \Delta\mathbf{l}_{1-2} = \Delta y \hat{\mathbf{a}}_y$$

$$\lim_{\Delta S_x \to 0} \frac{\oint \mathbf{H} \cdot d\mathbf{l}}{\Delta S_x}\hat{\mathbf{a}}_x = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right)\hat{\mathbf{a}}_x = J_x \hat{\mathbf{a}}_x$$

$$\nabla \times \mathbf{H} = \begin{vmatrix} \hat{\mathbf{a}}_x & \hat{\mathbf{a}}_y & \hat{\mathbf{a}}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$ ระบบพิกัดฉาก

$$\nabla \times \mathbf{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \hat{\mathbf{a}}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \hat{\mathbf{a}}_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \hat{\mathbf{a}}_z$$

ระบบพิกัดฉาก

$$\nabla \times \mathbf{H} = \left(\frac{1}{\rho} \frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z} \right) \hat{\mathbf{a}}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho} \right) \hat{\mathbf{a}}_\phi + \frac{1}{\rho} \left(\frac{\partial (\rho H_\phi)}{\partial \rho} - \frac{\partial H_\rho}{\partial \phi} \right) \hat{\mathbf{a}}_z$$

ระบบพิกัดทรงกระบอก

$$\nabla \times \mathbf{H} = \frac{1}{r \sin\theta} \left(\frac{\partial (H_\phi \sin\theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \hat{\mathbf{a}}_r + \frac{1}{r} \left(\frac{1}{\sin\theta} \frac{\partial H_r}{\partial \phi} - \frac{\partial (r H_\phi)}{\partial r} \right) \hat{\mathbf{a}}_\theta + \frac{1}{r} \left(\frac{\partial (r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right) \hat{\mathbf{a}}_\phi$$ ระบบพิกัดทรงกลม

ฟลักซ์แม่เหล็ก และความหนาแน่นฟลักซ์แม่เหล็ก (magnetic flux and magnetic flux density)

$$\mathbf{B} = \mu \mathbf{H},\ \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

กฎของฟาราเดย์ (Faraday's law)

$$\text{emf} = -\frac{d\Phi}{dt},\ \text{emf} = \oint \mathbf{E} \cdot d\mathbf{l},\ \text{emf} = \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_S (\mathbf{B} \cdot d\mathbf{s})$$

$$\oint \mathbf{E} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s},\ \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s},\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial}{\partial t} \int_S (\mathbf{B} \cdot d\mathbf{s})$$

กระแสไฟฟ้าดิสเพลสเมนท์ (displacement electric current)

$$\nabla \times \mathbf{H} = \mathbf{J} + \mathbf{G},\ \nabla \cdot (\nabla \times \mathbf{H}) = \nabla \cdot \mathbf{J} + \nabla \cdot \mathbf{G},\ \nabla \cdot (\nabla \times \mathbf{H}) = 0,\ \nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

$$\mathbf{G} = \frac{\partial \mathbf{D}}{\partial t},\ \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t},\ \mathbf{J}_d \equiv \frac{\partial \mathbf{D}}{\partial t},\ \nabla \times \mathbf{H} = \mathbf{J} + \mathbf{J}_d$$

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} = \int_S \mathbf{J} \cdot d\mathbf{s} + \int_S \mathbf{J}_d \cdot d\mathbf{s}$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{s} + \int_S \mathbf{J}_d \cdot d\mathbf{s},\ \oint \mathbf{H} \cdot d\mathbf{l} = I + I_d,\ I \equiv \int_S \mathbf{J} \cdot d\mathbf{s},\ I_d \equiv \int_S \mathbf{J}_d \cdot d\mathbf{s}$$

รูปแบบจุดของชุดสมการของแมกซ์เวลล์ (point form of Maxwell's equations)

$$\nabla \cdot \mathbf{D} = \rho_v,\ \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t},\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},\ \nabla \cdot \mathbf{B} = 0$$

รูปแบบปริพันธ์ของชุดสมการของแมกซ์เวลล์ (integral form of Maxwell's equations)

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_{\text{vol}} \rho_v dv,\ \oint \mathbf{H} \cdot d\mathbf{l} = I + \frac{\partial}{\partial t}\int_S (\mathbf{D} \cdot d\mathbf{s}),\ \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial}{\partial t}\int_S (\mathbf{B} \cdot d\mathbf{s})$$

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$$

ความหนาแน่นกระแสไฟฟ้าการนำ (conduction electric current density)

$$\mathbf{J} = \sigma \mathbf{E}$$

ความหนาแน่นกระแสไฟฟ้าการพา (convection electric current density)

$$\mathbf{J} = \rho_v \mathbf{v}$$

เงื่อนไขขอบเขต (boundary conditions)

$$E_{t1} = E_{t2},\ H_{t1} = H_{t2},\ D_{N1} - D_{N2} = \rho_s,\ B_{N1} = B_{N2}$$

ชุดสมการของแมกซ์เวลล์ในรูปแบบฮาร์มอนิกเวลา
(Maxwell's equations in time-harmonic form)
และคลื่นระนาบสม่ำเสมอ (uniform plane wave)

$$\mathbf{J} = 0,\ \rho_v = 0,\ \nabla \cdot \mathbf{E} = 0,\ \nabla \cdot \mathbf{H} = 0$$

$$\nabla \times \mathbf{H}_s = j\omega\varepsilon_0 \mathbf{E}_s,\ \nabla \times \mathbf{E}_s = -j\omega\mu_0 \mathbf{H}_s$$

$$\nabla \times \mathbf{E}_s = \begin{vmatrix} \hat{\mathbf{a}}_x & \hat{\mathbf{a}}_y & \hat{\mathbf{a}}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ E_{xs} & E_{ys} & E_{zs} \end{vmatrix}$$

ระบบพิกัดฉาก

$$\nabla \times \mathbf{E}_s = \left(\frac{\partial E_{zs}}{\partial y} - \frac{\partial E_{ys}}{\partial z}\right)\hat{\mathbf{a}}_x + \left(\frac{\partial E_{xs}}{\partial z} - \frac{\partial E_{zs}}{\partial x}\right)\hat{\mathbf{a}}_y + \left(\frac{\partial E_{ys}}{\partial x} - \frac{\partial E_{xs}}{\partial y}\right)\hat{\mathbf{a}}_z$$

ระบบพิกัดฉาก

$$\nabla \times \nabla \times \mathbf{E}_s = -j\omega\mu_0 \nabla \times \mathbf{H}_s$$

$$\nabla \times \nabla \times \mathbf{E}_s = \nabla(\nabla \cdot \mathbf{E}_s) - \nabla^2 \mathbf{E}_s,\ \nabla \times \nabla \times \mathbf{E}_s = -\nabla^2 \mathbf{E}_s$$

$$\nabla^2 \mathbf{E}_s = -\omega^2 \mu_0 \varepsilon_0 \mathbf{E}_s,\ \nabla^2 \mathbf{E}_s = -k_0^2 \mathbf{E}_s,\ k_0 = \omega\sqrt{\mu_0 \varepsilon_0}$$

$$\mathbf{E}_s = E_{xs}(z)\hat{\mathbf{a}}_x,\ \nabla^2(E_{xs}\hat{\mathbf{a}}_x) = -k_0^2 E_{xs}\hat{\mathbf{a}}_x,\ \frac{\partial^2 E_{xs}}{\partial x^2} + \frac{\partial^2 E_{xs}}{\partial y^2} + \frac{\partial^2 E_{xs}}{\partial z^2} = -k_0^2 E_{xs}$$

$$(\omega t - k_0 z) = \text{constant},\ \omega dt - k_0 dz = 0,\ \frac{dz}{dt} = v = \frac{\omega}{k_0},\ v = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = c$$

$$c = 3\times 10^8,\ \varepsilon_0 = 8.854\times 10^{-12}\ \text{F/m},\ \mu_0 = 4\pi\times 10^{-7}\ \text{H/m}$$

$$\eta_0 \equiv \sqrt{\frac{\mu_0}{\varepsilon_0}} = 377 = 120\pi$$
